

GPT Working

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How GPT Generates Content?

Artificial Intelligence (AI) is one of the most significant technological advancements of the modern world. It refers to the ability of machines and computer systems to perform tasks that usually require human intelligence. These tasks include learning from experience, solving problems, understanding language, recognizing images, making decisions, and even creating new content. AI has become an essential part of our daily lives, transforming the way we work, communicate, and solve complex problems.

<----- I asked ChatGPT to write an essay for me

Working Mechanism is like:

At time step $t = 0$

Artificial ---> model --> Intelligence

At time step $t = 1$

Artificial Intelligence ---> **model** --> (

At time step $t = 2$

Artificial Intelligence (---> **model** --> AI

At time step $t = 3$

Artificial Intelligence (AI ---> **model** -->)

At time step $t = 4$

Artificial Intelligence (AI) ---> **model** --> is

At time step $t = 5$

Artificial Intelligence (AI) is ---> **model** --> one

At time step $t = 6$

Artificial Intelligence (AI) is one ---> **model** --> of

At time step $t = 7$

Artificial Intelligence (AI) is one of ---> **model** --> the

At time step $t = 8$

Artificial Intelligence (AI) is one of the ---> **model** --> most

At time step $t = 9$

Artificial Intelligence (AI) is one of the most ---> **model** --> significant

.
. .
. .
. .
. .

At time step $t = t_n$

Artificial Intelligence (AI) is one of the most significant technological advancements of the modern world. It refers to the ability of machines and computer systems to perform tasks that usually require human intelligence. -----> **Model** -----> These

Problem: At every time step $t = t_n$ the model needs all the previous tokens to predict the next token
As the sentence becomes longer the memory required also increases

Solution: Instead of passing the entire sentence we only pass the latest context using a Context Window

Context Window: A Context Window is the maximum number of previous tokens that the model can look at while predicting the next token

Instead of remembering the entire sentence the model only keeps the most recent tokens in its working memory

For this example let us assume the context window size is 10 tokens

[, , , , , , , , , ,] <----- **Context Window**

Now let's see the same example but with integrated context window

Working Mechanism is like:

At time step $t = 0$

[Artificial,] --> model --> Intelligence

At time step $t = 1$

[Artificial, Intelligence] --> **model** --> (

At time step $t = 2$

[Artificial, Intelligence, (] --> **model** --> AI

At time step $t = 3$

[Artificial, Intelligence, (, AI] --> **model** -->)

At time step $t = 4$

[Artificial, Intelligence, (, AI,)] --> **model** --> is

At time step $t = 5$

[Artificial, Intelligence, (, AI,), is] --> **model** --> one

At time step $t = 6$

[Artificial, Intelligence, (, AI,), is, one] --> **model** --> of

At time step $t = 7$

[Artificial, Intelligence, (, AI,), is, one, of] --> **model** --> the

At time step $t = 8$

[Artificial, Intelligence, (, AI,), is, one, of, the] --> **model** --> most

At time step $t = 9$

[Artificial, Intelligence, (, AI,), is, one, of, the, most] --> **model** --> significant

At time step $t = 10$

[Intelligence, (, AI,), is, one, of, the, most, significant] --> **model** --> technological

At time step $t = 11$

[(, AI,), is, one, of, the, most, significant, technological] --> **model** --> advancements

At time step $t = 12$

[(, AI,), is, one, of, the, most, significant, technological, advancements] --> **model** --> of

At time step $t = 13$

[AI,), is, one, of, the, most, significant, technological, advancements, of] --> **model** --> the

At time step $t = 14$

[), is, one, of, the, most, significant, technological, advancements, of, the] --> **model** --> modern

.
. .
. .
. .
. .

At time step $t = t_n$

[systems, to, perform, tasks, that, usually, require, human, intelligence, .] ----->

Model -----> These

Notice: As the sentence grows we stop passing the entire sentence to the model.
Instead we only pass the latest 10 tokens for predicting the next token

In practice modern GPT models use much larger context windows than 10 tokens

Biggest Question

How do we predict next token?

Answer: We use Attention Mechanism to do so.

Example:

The animal didn't cross the road because it was tired. --> [here 'it' refers to the "Animal"] --> we have total of 10 tokens

Imagine you're reading a sentence. When you see the word "it", your brain automatically searches for which earlier word "it" refers to. GPT does the same thing mathematically using Query, Key, and Value vectors.

GPT: Uses this same Attention Mechanism

This Attention mechanism internally uses this:

1. Each token will maintain set of 3 vectors called as Query, Key and Value vectors
2. And we use this Vectors to pay attention to some specific tokens whenever we want
 - a. **Query:** What am I looking for?
 - b. **Key:** Contains information of what I can contribute to other tokens
 - c. **Value:** This is the information that I will contribute if you will attend to me

The --> Query = [0.7, -0.9], Key = [1.2, -0.4], Value = [0.8, 0.3]

animal --> Query = [-0.4, 1.6], Key = [0.9, -1.4], Value = [6.5, -8.3]

didn't --> Query = [1.1, 0.2], Key = [-0.8, 0.5], Value = [0.4, -1.0]

cross --> Query = [-1.3, 0.8], Key = [0.6, 1.0], Value = [-0.7, 1.5]

the --> Query = [0.5, -1.1], Key = [-1.2, -0.3], Value = [1.2, 0.7]

road --> Query = [0.9, 0.6], Key = [0.3, -0.9], Value = [-0.4, 0.8]
because --> Query = [-0.8, -0.5], Key = [1.1, 0.4], Value = [0.6, -0.2]
it --> Query = [1.4, -0.7], Key = [0.8, 1.3], Value = [-1.1, 0.5]
was --> Query = [-0.2, 1.0], Key = [-0.5, 0.9], Value = [0.3, 1.1]
tired --> Query = [0.6, -0.4], Key = [1.4, -1.0], Value = [0.9, -0.6]

Attention Definition:

- $\text{Softmax}(\text{Query} @ \text{Key.T}) @ \text{Value}$

NOTE: I'm Intentionally ignoring the Scaling Factor $1/\sqrt{d_k}$

One example of this is:

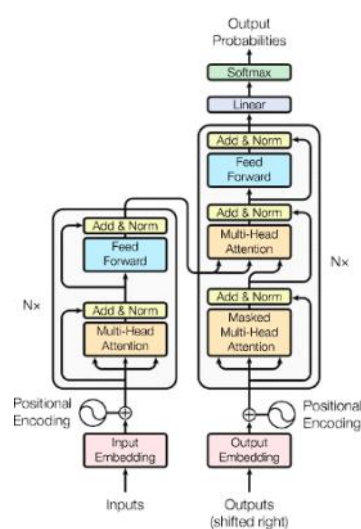
- My name is Joanna
 - o My: Q = [0.3, 0.4]; K = [-1.4, -0.6], V = [0.4, 0.6]
 - o Name: Q = [0.8, -0.5]; K = [0.2, 1.1]; V = [-0.3, 0.9]
 - o is: Q = [-0.6, 1.2]; K = [0.7, -0.8]; V = [1.0, -0.4]
 - o Joanna: Q = [1.1, 0.2]; K = [-0.9, 0.5]; V = [0.7, -1.2]

----- Working Example of Attention Mechanism -----

While generative new tokens, the tokens can now attend to each other as:

- How much information does 'My' is contributing to 'Joanna'
 - o Joanna token will raise a Query and it will attend to token 'My'
 - $(\text{Softmax}(Q_{\text{joanna}} @ [K_{\text{my}}, K_{\text{name}}, K_{\text{is}}, K_{\text{Joanna}}])) @ [V_{\text{my}}, V_{\text{name}}, V_{\text{is}}, V_{\text{Joanna}}]$
 - $Q_{\text{Joanna}} @ K_{\text{my}} = (1.1 * -1.4 + 0.2 * (-0.6)) = -1.66$
 - $Q_{\text{Joanna}} @ K_{\text{name}} = (1.1 * 0.2) + (0.2 * 1.1) = 0.44$
 - $Q_{\text{Joanna}} @ K_{\text{is}} = (1.1 * 0.7) + (0.2 * (-0.8)) = 0.61$
 - $Q_{\text{Joanna}} @ K_{\text{Joanna}} = (1.1 * 0.9) + (0.2 * 0.5) = -0.89$
 - Attention Scores are: [-1.66, 0.44, 0.61, -0.89]
 - $\text{Softmax}(Q_{\text{joanna}} @ [K_{\text{my}}, K_{\text{name}}, K_{\text{is}}, K_{\text{Joanna}}]) = [0.048, 0.389, 0.461, 0.103]$
 - Now multiply with Value vectors:
 - ◆ Contribution from 'My': $0.048 \times [0.4, 0.6] = [0.019, 0.029]$
 - ◆ Contribution from 'Name': $0.389 \times [-0.3, 0.9] = [0.117, 0.350]$
 - ◆ Contribution from 'is': $0.461 \times [1.0, -0.4] = [0.461, -0.184]$
 - ◆ Contribution from 'Joanna': $0.103 \times [0.7, -1.2] = [0.072, -0.124]$
 - Add All contributions
 - ◆ $[0.019, 0.029] + [0.117, 0.350] + [0.461, -0.184] + [0.072, -0.124] = [0.435, 0.071]$
 - o Now this final output will be passed in the MLP (Multi-Layer Perceptron) for further analysis and it will encode more useful representation

The Entire Architecture: Input --> (Position + Word Embeddings) --> Self-Attention (what we saw) --> MLP --> Prediction; This is exactly what Transformer Block is



The Entire Architecture (Transformer)